



**Amrita School of Artificial
Intelligence**

COIMBATORE CAMPUS

AI IN MATERIAL SCIENCE

Reading List

Where to go next after the AI in Material Science workshop

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WORKSHOP

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FROM

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COST

Everything listed here is free

OFFERING

University of Calicut, 24 September 2026

Start here

Today's notebooks and data. Everything you ran this afternoon, including the two notebooks, the 1128-molecule solubility dataset and the setup instructions. Change the CSV to your own measurements and the same code works. Ask me for the link if you do not have the USB stick.

This workshop. github.com/Amrita-School-of-AI/ai-in-material-science The notebooks you ran today, the slides, the data and the morning cells. Clone it and change the dataset to your own.

CHEM 5080, AI for Experimental Chemistry. zzhenglab.github.io/ai4chem Zhiling Zheng's course at Washington University in St. Louis. Twenty-five lectures, written for synthetic chemists with no programming background, every one of them a notebook you can open in Google Colab and run. Start at lecture 1 even if today felt comfortable, and you will reach graph neural networks by lecture 9. This is the single best thing on this page. The course paper is Zheng, "Developing an AI Course for Synthetic Chemistry Students", *Journal of Chemical Education* 103, 2621 (2026).

RDKit documentation. rdkit.org/docs The toolkit behind everything in Hands-on 1. Read "Getting Started with RDKit in Python" once, then keep the Cookbook open while you work.

Go deeper

Delaney, "ESOL: Estimating Aqueous Solubility Directly from Molecular Structure", *J. Chem. Inf. Comput. Sci.* 44, 1000 (2004). The source of the dataset you fitted this afternoon. Short, readable, and a good model of how to write up a structure-property study.

Stokes et al., "A Deep Learning Approach to Antibiotic Discovery", *Cell* 180, 688 (2020). The halicin paper. Worth reading in full because the machine learning is only half of it; the other half is the experimental validation, which is what made it matter.

Butler et al., "Machine learning for molecular and materials science", *Nature* 559, 547 (2018). A readable overview of the field written for chemists rather than computer scientists. Dated in its examples now, still correct in its framing.

Bran et al., "Augmenting large language models with chemistry tools", *Nature Machine Intelligence* 6, 525 (2024). ChemCrow. The full version of the agent you built in Hands-on 2, with eighteen tools and real syntheses at the end of it.

The tools

scikit-learn. scikit-learn.org The machine-learning library used this afternoon. Its user guide is a genuine textbook, not just an API reference.

Chemprop. github.com/chemprop/chemprop Graph neural networks for molecular property prediction, from MIT. When descriptors stop being enough, this is the next step, and it runs from the command line without writing a model yourself.

Materials Project and pymatgen. materialsproject.org and pymatgen.org Computed properties for well over a hundred thousand inorganic crystals, free with registration, plus the Python library to work with them. The starting point for anyone on the materials side rather than the molecules side.

MACE. github.com/ACEsuit/mace Machine-learned interatomic potentials with pretrained models you can download and run. If you do molecular dynamics, this is the change worth making.

ChemMatData. From [aimat-lab](https://aimat-lab.org), an overview of chemical and materials datasets and where to find them. Useful when your problem is that you have no data.

aimat-lab. github.com/aimat-lab The AI for Materials group at KIT Karlsruhe led by Pascal Friederich. Open code and datasets, a good picture of what a working research group in this area actually produces.

The agent frameworks

ChemGraph. github.com/argonne-lcf/ChemGraph From Argonne National Laboratory. The research-grade version of Hands-on 2: a natural-language request becomes a real quantum chemistry or machine-learned-potential calculation. Apache licensed. Paper: Pham, Tanikanti and Keçeli, *Communications Chemistry* 9, 33 (2026).

Scientific agent skills. github.com/k-dense-ai/scientific-agent-skills A large collection of ready-made skills that give a coding agent working knowledge of RDKit, DeepChem, pymatgen and around a hundred scientific databases. MIT licensed.

LangGraph. langchain-ai.github.io/langgraph The library behind the agent loop you ran. The quickstart is short and the concepts map directly onto what you saw this afternoon.

If something here does not work, or you want to try this on data from your own laboratory, write to me at a_abhijith@cb.amrita.edu. Tell me what you are trying to predict and what data you have, and I will point you at the right starting notebook.

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